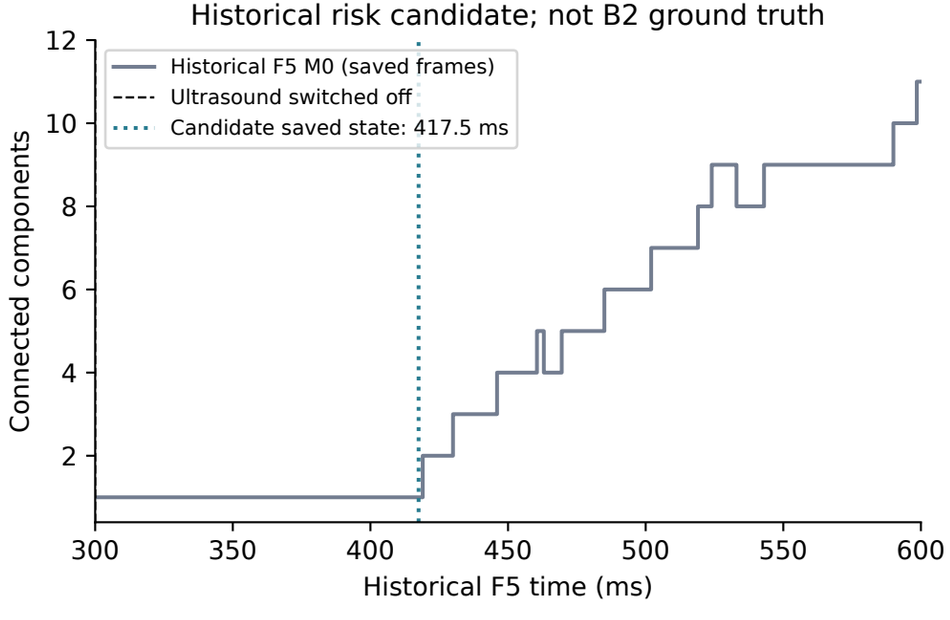
Fixed-state numerical admission and limits of scientific coverage

Scope	Measured evidence	Decision	Limit
Two-bubble references	12 / 12 matched	**PASS**	Original equation and forces
Four-bubble references	4 accepted; 2 budget negatives	**PASS**	Both over-budget states rejected
Real 32-bubble state	1.107e-16 m residual	**PASS**	Old 1e-14 m gradient criterion met
Contact budgets	17.498 nm; 1.932 nm	**PASS**	Overlap ≤17.8 nm; one correction ≤178 nm
GPU implementation	Corrected / optimized positions identical	**PASS**	Fixed-state tests only
Fair B = 1 performance	GPU 22.05 / CPU 11.87 ms	**NO ADVANTAGE**	Stop production integration
Contact / near-field physics	Exploratory stiffness and approximations	**UNCALIBRATED**	Not experimental truth
AI data admission	training_ready = false	**NOT ADMITTED**	No new dataset or training



PERSISTENT TOPOLOGY VALIDATION NOT RUN

98 historical ledgers reconstructed; original 2 ms dwell.
F5 release: 11 persistent splits, 2 merges.
Candidate needs B2 restart and near-wall validation.

F10H/I geometric B2: one connected group throughout.
No observed persistent split or merge to validate.
Original strict edge-time failure remains: 12 / 29.

Group position / shape evidence is retained.
No new topology trajectory, B64 run, or AI training.
Decision: preserve CPU reference; stop GPU production route.

EXPLORATORY / PHYSICS_UNCALIBRATED. Numerical fixed-state agreement does not calibrate contact physics or validate persistent topology.
Historical runs are correlated evidence, not independent scientific samples. Original F10I failures and all quality budgets are retained.